

□ C++ Intermediate Assessment

Test Reference No: PLMAcademy- Nov2025-General-Batch

Total Marks : 30

Test Duration : 1 hour

Instructions to Candidate

1. All questions are compulsory/mandatory.
2. Do not write anything on the question paper.
Write everything on the answer sheet provided.
3. This is a closed book test.
4. There is no negative marking.
5. Wherever there is ambiguity, make suitable assumptions and proceed.
Document the assumptions in the answer.
6. Assume C++ 11 language standard.
7. Assume usage of Microsoft Visual Studio development platform.

Question No.	Question	Marks
1	Show how will you call the static member function of a class? <pre>#include <iostream> using namespace std ; class X { public: static void Show() { cout << "X::Show() called" << endl ; } }; int main () { // Write code which calls the static function of class X. return 0 ; }</pre>	1
2	A reference refers to a referent". Write a small piece of code, along with memory diagram which explains/demonstrates this statement. Identify the reference and referent in your code.	2
3	Will the below code compile and execute successfully. If not mention : 1. Whether the error is compile time or runtime?	2

	2. What that error is? <pre> class X { public: X(int i) { m_i = i; } private: int m_i; }; class Y : public X { public: Y(); }; Y::Y() { } int main() { Y y; return 0; } </pre>	
4	Write a program which creates a global object of class X on heap. <pre> class X { }; </pre>	1
5	What will be the output of the following code? <pre> class X { public: static int m_i; }; int main () { cout << sizeof(X) << endl; return 0; } </pre>	1

6	What is an abstract class? Explain with minimal example.	2
7	Will the following code compile and execute successfully? If not mention which line number will give the error and what error ,it will be? <pre> 1 #include <iostream> 2 using namespace std; 3 int main() { 4 int x=10, int y=20; 5 int* p1, p2; 6 p1 = &x; 7 p2 = &y; 8 return 9 }</pre>	1
8	What will be the output of the following code: <pre> #include <iostream> using namespace std; int main() { char s[] = "abcd"; char* p = s; *p++ = 'z'; ++*p; cout << s; return 0; }</pre>	1
9	What does the new and delete operator do? Explain with minimal code and memory diagram.	2
10	Will the code compile? If not fix the code so that it will be successfully compiled and executed. Note: Destructor needs to be pure virtual only. <pre> #include <iostream> using namespace std;</pre>	2

	<pre> class Shape { public: void setData() {} virtual ~Shape() = 0; }; class Circle : public Shape { }; int main() { Shape *x=new Circle(); delete x; return 0; } </pre>	
11	Will the below code compile and execute successfully? If so ,what will be the output of the following code? <pre> #include <iostream> using namespace std; int main() { char str[] = "pointers"; char* p = str; cout << *(p + 2); cout << p[2]; cout << 2[p]; }all </pre>	1
12	Why should a function not return the address of a local non-static variable? Explain with minimal code and memory diagram.	2
13	What is wrong in the following code: <pre> #include <iostream> using namespace std ; int i = 100 ; </pre>	2

	<pre> int & f() { return i ; } int main() { f() = 200 ; cout << i << endl ; return 0 ; } </pre>	
14	What will be the output of the following program? <pre> #include <iostream> using namespace std ; int main () { int i = 20 ; const int *pi = &i ; *pi = 200 ; cout << i << endl ; return 0 ; } </pre>	1
15	What will be the output of the following program: <pre> #include <iostream> using namespace std; class Shape { int x; public: Shape(int y):x(y){} Shape(const Shape& s) { cout << "Shape Copy constructor called"; x = s.x; } }; int main() { Shape x(10), z(20); Shape y = x; } </pre>	1

	<pre> z = y; return 0; } </pre>	
16	What is the command to compile the file a.cpp, into an a.obj. What is the command to get the symbol table from an object file? What will be the important/significant symbol table entries for the following code with respect to the functions in the following code: <pre> void f (); void g() { } int main () { f (); g (); } File a.cpp </pre>	2
17	Based on the code below, answer the following: 1. How many VTable will be created for the below code? 2. How many VPTR are created for the below code? 3. What is the output of the below code? <pre> #include <iostream> using namespace std; #pragma pack(1) class Shape { int x; public: virtual void f(){} void g(){} }; class Circle : public Shape{ </pre>	3

	<pre> int y; public: void f(){} }; int main() { Shape* s = new Circle(); cout << "Size of Shape class " << sizeof(Shape) << endl; cout << "Size of Circle class " << sizeof(Circle) << endl; return 0; } </pre>	
18	Why does the following code not compile? <pre> #include <iostream> using namespace std; int main() { int enemy = 0; // number of enemies. int friend = 0; // number of friends. return 0; } </pre>	1
19	Below function max returns the maximum of the two integers. Write an expression in the return statement to find out the maximum, and return the maximum value. <pre> int max (int i, int j) { return (_____); } </pre>	1
20	What will be the output of the following code: <pre> #include <iostream> using namespace std; class CEmployee { int ID; public: ~CEmployee() { cout << "Destructor called" << endl; } } </pre>	1

	<pre>CEmployee() { ID = 1; } CEmployee(int x):ID(x){ void setID(int x) { ID = x; } int getID() { return ID; } }; class CEngineer : public CEmployee { public: CEngineer(int x) { setID(x); } int getSAPID() { return CEmployee::getID(); } }; int main() { CEmployee* e = new CEngineer(1001); cout << e->getID() << endl; }</pre>	
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